

Response to Reviewers

Manuscript: *Matryoshka Representation Learning for Adaptive-Fidelity ECG Classification*

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Summary of changes

We thank both reviewers for reviews that identified real weaknesses rather than cosmetic ones. Two criticisms in particular changed how we understand our own result, and we have restructured the paper accordingly rather than patching around them.

1. **The wearable-to-cloud framing was not supported and has been withdrawn.** The paper now claims adaptive embedding compression on clinical twelve-lead ECG. The device-tier mapping is presented as arithmetic, not as a validated deployment result, and the title, abstract, introduction and conclusion have been rewritten accordingly. We have added reduced-lead and artefact-corruption experiments that bound the wearable question without pretending to settle it.
2. **The computational argument was wrong in kind, not merely in precision.** Reviewer 1’s observation that the backbone dominates inference cost is correct and is fundamental to the method: prefix truncation occurs after the encoder has run, so latency is flat in d by construction. We now measure this on real hardware and state it explicitly as a limitation of MRL. The previous device-tier latency table, produced by scaling one GPU measurement by a ratio of nominal device throughputs, has been deleted.
3. **Fixed-dimension baselines with the same backbone have been added** at every nesting dimension and every seed. We further note that our previous headline comparison—an Inception1D MRL model against XResNet1D-101 fixed baselines—confounded nesting with backbone choice and could not support the conclusion we drew from it.
4. **Uncertainty quantification throughout:** five seeds per condition, bootstrap confidence intervals, DeLong tests, Holm correction, and an equivalence test with a pre-registered margin for the central invariance claim.
5. **The clinical feature hierarchy is now measured**, by ridge probing of eleven physiological descriptors against each embedding prefix. We commit to reporting the outcome whichever way it falls, including withdrawing the hierarchy table if the probes do not support it.
6. **Cross-dataset evaluation** on CPSC-2018, Chapman-Shaoxing and Georgia, with an explicit statement that the SNOMED-CT to superclass mapping is lossy.

We also corrected four issues the reviewers did not raise but which we found while preparing the revision, listed in Section 3.

Note to co-authors: numbers marked [TBD] in the revised manuscript are populated automatically from `scripts/aggregate_results.py` once the experiment matrix has been run. This letter should be finalised after that point, and the outcome-dependent passages below (marked †) resolved to what was actually measured.

1. Reviewer 1

Major comment 1 — Wearable claim not supported by PTB-XL

Comment. *The evaluation is performed only on PTB-XL... substantially different from wearable ECGs, ambulatory recordings, noisy signals, and single-lead settings... Without one of these additions, the deployment claim should be recalibrated and framed more narrowly as adaptive embedding compression on PTB-XL.*

Response. We accept this in full, and have taken the recalibration option seriously rather than treating it as a fallback.

The title no longer contains “wearable-to-cloud”. Section I now includes an explicit subsection, *Scope, and what this paper does not claim*, separating three claims that the previous version conflated: embedding compressibility (demonstrated), deployment mapping (arithmetic), and on-device wearable validation (not established). The abstract states the last of these as an open question.

We have additionally added the strongest proxies available without new data collection:

- **Cross-dataset transfer** to CPSC-2018, Chapman–Shaoxing/Ningbo and Georgia (Section V-E1). We report these with the caveat that the SNOMED-CT to superclass mapping is many-to-one and lossy, so they measure concept transfer under shift rather than identical-task performance.
- **Reduced-lead evaluation** down to single-lead input, in both zero-shot and natively-trained regimes (Section V-E2).
- **Artefact robustness** across baseline wander, EMG, electrode motion and mains interference at SNRs from 20 dB to 0 dB (Section V-E3).

We are explicit that these are proxies: synthesised artefacts injected into clinically acquired signals are not dry-electrode wearable recordings, and masking leads is not a single-lead acquisition geometry. Regarding the suggestion of HEEDB, MIMIC-IV-ECG or CODE-15%, our released pipeline supports these corpora directly; we have not exercised them because they require credentialed access, and we identify scale as an explicit limitation (Section VI-D).

Changes: Title; abstract; new Section I-A; new Sections V-E1–V-E3; Figs. (leads, robustness); Table (external); Section VI-B, VI-D.

Major comment 2a — Computational benefit not established

Comment. *Reducing the embedding dimensionality reduces representation size, but the full backbone still has to be executed and appears to dominate inference cost. Therefore, claims about latency, memory, and energy advantages require real-device benchmarking or a more careful limitation of the claims.*

Response. This is the most important comment in either review, and the reviewer is right about the mechanism, not merely about the evidence. Prefix truncation is a pointer offset applied *after* the encoder has run; latency, energy and peak activation memory are therefore invariant to d by construction. No amount of additional measurement would change this.

We have done both things the reviewer offers as alternatives. We now measure on real hardware—GPU with CUDA-event timing and NVML energy integration, multi- and single-threaded CPU, dynamically quantised CPU, and ONNX Runtime—and we have rewritten the efficiency claim. Section V-C now states that nesting reduces (i) embedding storage and transmission, by up to 32 $\times$, and (ii) the number of maintained model artefacts, with the associated validation and certification burden; and that it does *not* reduce inference latency, energy or

activation memory. Table (related work) now carries a “compute saving” column in which our own method is marked *No*.

The previous Table III has been deleted. Its Cortex-M4 column was an analytical extrapolation from an Apple MPS measurement scaled by nominal device throughput, which we should not have presented in a table of measured quantities.

Changes: Section V-C rewritten; new measured Table (compute) and Fig. (latency); old Table III removed; Section VI-B; related-work table column added; `scripts/benchmark_hardware.py` released.

Major comment 2b — Fixed-dimension Inception1D baselines

Comment. *Since the best proposed model uses Inception1D, fixed-dimension Inception1D baselines should be included.*

Response. Added, at all six dimensions and all five seeds, for both backbones.

We want to acknowledge the deeper issue this exposes. Our previous headline claim—that the MRL model “outperforms six independently trained 34M-parameter baselines”—compared an Inception1D MRL model against XResNet1D-101 fixed baselines. Since Inception1D also outperformed XResNet1D-101 *under MRL*, that comparison confounded nesting with backbone choice and could not establish what we claimed. The matched baselines now separate the two factors, and we have removed the inference that the multi-granularity loss acts as a regulariser, which rested on the same confounded comparison (Section VI-C).

Changes: Table (main) restructured; Section V-B; Section VI-C; `run_all.sh` Stage 3.

Major comment 2c — Repeated runs and statistical testing

Comment. *Repeated runs, confidence intervals, or statistical testing are also needed, because some reported differences are small.*

Response. Every condition is now trained with five seeds and reported as mean $\pm$ SD. Test-set uncertainty uses bootstrap over recordings ($B=2,000$); model comparisons use paired bootstrap and DeLong’s test with Holm correction across the six dimensions.

One methodological point we would highlight: our central claim is that performance is *stable* across dimensions, which is a claim of no meaningful difference. A non-significant superiority test cannot establish that. We therefore pre-register an equivalence margin of ± 0.01 macro AUC (roughly half the seed-to-seed SD of a PTB-XL classifier) and require the whole confidence interval of the difference to fall inside it (Section IV-E, Table (stats)).

Changes: New Section IV-E; Table (stats); all tables report SD; seed bands in all figures; `mecg/analysis/stats.py`.

Minor comment 1 — Terminology

Comment. *The terminology around “wearable”, “IoT”, and “deployment” should be made more precise, distinguishing between conceptual deployment mapping, actual on-device inference, and validation on wearable data.*

Response. We have adopted exactly this three-way distinction and made it structural: it is enumerated in Section I-A, maintained consistently in Sections III-F, V-C and V-E, and revisited in the discussion. “IoT” has largely been removed, since it was doing rhetorical rather than technical work. We now write “deployment mapping” for the arithmetic, “on-device inference” for execution on target hardware, and “wearable validation” for evaluation on wearable-acquired data, and we state which of the three each claim belongs to.

2. Reviewer 2

Comment 1 — Does not outperform stronger PTB-XL classifiers

Comment. *The proposed model does not outperform stronger reported PTB-XL classifiers and its contribution is mainly adaptive compression rather than improved ECG diagnosis.*

Response. Correct, and we now say so directly rather than allowing the framing to imply otherwise. Section V-G presents the comparison with an explicit statement that we make no accuracy claim, and that published non-adaptive systems each serve one operating point. Section I states that we decline the superiority framing, and the abstract characterises the contribution as concerning the *shape* of the accuracy–dimension curve rather than its height.

We also note—since it bears on the reviewer’s point—that our absolute performance is below the Strodthoff *et al.* benchmark for the same architectures. We identified a probable cause during revision (Section 3, item 1) and have corrected it; whether the gap closes will be evident in the revised results.

Comment 2 — Missing fixed Inception1D baseline

Response. Addressed above (Reviewer 1, major 2b).

Comment 3 — Deployment claims not validated on hardware

Response. Addressed above (Reviewer 1, major 2a). Measurements are now real; the claim has been narrowed to what measurement supports.

Comment 4 — Clinical hierarchy is speculative

Comment. *The proposed clinical hierarchy of embedding dimensions is speculative and not validated by probing or interpretability analysis.*

Response. Accepted. The previous Table II asserted a mapping from nesting dimensions to clinical features on the basis of a gradient-flow argument that establishes only that leading coordinates receive more gradient signal—not that they encode any *particular* feature. The inference did not follow.

We now test it. Eleven physiological descriptors (RR statistics, QRS duration, QT, ST level, T-wave amplitude, R and P amplitude, frontal axis) are extracted from the raw waveforms; ridge probes are fitted on training-split embeddings and evaluated out-of-sample; and we record the saturation dimension for each descriptor. Section V-D reports the rank correlation between hypothesised and measured ordering, plus prefix CKA, effective rank and the variance profile.

† We state in the manuscript, before seeing the outcome, what each result would mean: if saturation order tracks the hypothesis, the nesting structure has a physiological interpretation; if all descriptors saturate at the smallest prefix, the parsimonious conclusion is that the diagnostic manifold is simply low-dimensional, which explains the flat AUC profile equally well but licenses no claim about what particular coordinate ranges encode. In that case the hierarchy table will be withdrawn rather than reinterpreted.

Changes: Old Table II removed pending the probing result; new Section V-D; Table (probe); Fig. (probing); `mecg/analysis/probing.py`, `features.py`.

Comment 5 — Limited to PTB-XL five-class

Response. Cross-dataset evaluation added (Reviewer 1, major 1). On finer granularity: we have not added the 44-class task, and we explain why this is more than a resource question. If the flat profile is explained by low intrinsic dimensionality, the sufficient prefix should *grow* with

task granularity—so the 44-class result is a genuine test of the mechanism rather than a routine extension, and we would rather flag it as the sharpest open prediction (Section VI-D, VI-E) than report it hastily. Our preprocessing pipeline supports it via `--task subdiagnostic`.

Comment 6 — No uncertainty analysis

Response. Addressed above (Reviewer 1, major 2c).

3. Issues we identified ourselves

1. **Normalisation destroyed voltage information.** The original pipeline applied per-record, per-lead z -scoring, dividing each lead by its own standard deviation. This removes *absolute* amplitude—which is the diagnostic criterion for ventricular hypertrophy (Sokolow–Lyon, Cornell). HYP was by far our worst class (AUC 0.811 against 0.93 for NORM), and we believe this is why. Normalisation now uses training-split statistics; the comparison is retained as an ablation. This may also account for part of the gap to the published PTB-XL benchmark noted by Reviewer 2.
2. **Weight decay applied to normalisation parameters.** The optimiser grouped parameters by the substring test `'bn' in name`, which missed every BatchNorm nested inside a convolutional block. Now grouped by `param.ndim <= 1`.
3. **F1 evaluated at a fixed 0.5 threshold.** For an imbalanced multi-label task this substantially understates F1 and explains most of the apparent inconsistency between our AUC (≈ 0.90) and F1 (≈ 0.62). Thresholds are now tuned per class on validation and frozen for test.
4. **Paper–code mismatch and citation errors.** The manuscript described the Inception1D backbone as using squeeze-and-excitation attention; the released implementation did not. SE is now genuinely enabled. We also corrected citations: XResNet1D was attributed to He *et al.* (ResNet) and Inception1D to Szegedy *et al.* (GoogLeNet), where the relevant references are Strodtthoff *et al.* and Ismail Fawaz *et al.* (InceptionTime) respectively.
5. **SVD baseline leakage.** The basis and classifier were fitted on the test partition and evaluated on it. The manuscript acknowledged this as “optimistically biased” but nonetheless plotted the curve as an upper bound against honestly evaluated models. Both are now fitted on the training split.
6. **Model selection on the largest head.** Early stopping used validation AUC at $d=512$, biasing the checkpoint toward one operating point. Selection now uses the mean across granularities.
7. **Figure axis scaling.** The original Pareto figure used a 0.02-AUC vertical window, magnifying sub-0.005 differences into visually dramatic gaps and suggesting structure that is within seed noise. Axes are no longer zoomed and seed variability is shown.

Closing

Two of these criticisms—the compute mechanism and the confounded baseline—changed our understanding of what we had shown, and the revision is correspondingly a restructuring rather than an addition of experiments. We believe the narrower claim is the more useful contribution, and we are grateful to both reviewers for pressing on exactly the points that needed it.